

* Java Assignment 3

```
import java.util.*;
```

```
class InvalidMarksException extends Exception {  
    InvalidMarksException(String message) {  
        super(message);  
    }  
}
```

```
class Student {
```

```
    int rollNumber;
```

```
    String studentName;
```

```
    int[] marks = new int[3];
```

```
    public Student (int rollNumber, String studentName,  
                    int[] marks) {
```

```
        this.rollNumber = rollNumber;
```

```
        this.studentName = studentName;
```

```
        this.marks = marks;
```

```
    }
```

```
    void validateMarks() throws InvalidMarksException {
```

```
        int i = 0;
```

```
        while (i < marks.length) {
```

```
            if (marks[i] < 0 || marks[i] > 100) {
```

```
                throw new InvalidMarksException("invalid");
```

```
            }
```

```
            i++;
```

```
        }
```

```
    }
```

```
double calculateAverage() {
```

```
    int sum = 0;
```

```
    int i = 0;
```

```
    while(i < 3) {
```

```
        sum += marks[i];
```

```
        i++;
```

```
    }
```

```
    return sum/3.0;
```

```
}
```

```
void displayResults() {
```

```
    System.out.println("Name: " + StudentName);
```

```
    System.out.println("Roll: " + rollNumber);
```

```
    int i = 0;
```

```
    while(i < 3) {
```

```
        System.out.println("Marks " + (i+1) + ": " + marks[i]);
```

```
        i++;
```

```
    }
```

```
    double avg = calculateAverage();
```

```
    System.out.println("Average" + avg);
```

```
    if(avg >= 40) {
```

```
        System.out.println("Pass");
```

```
    } else {
```

```
        System.out.println("Fail");
```

```
    }
```

```
}
```

```
}
```

```
class ResultManager {
```

```
    Scanner input = new
```

```
    Student[] students
```

```
    int count = 0;
```

```
    void addStudent()
```

```
    try {
```

```
        System.out.println("Enter
```

```
        int roll
```

```
        input.next
```

```
        System.out.println("Enter
```

```
        String name
```

```
        int[] marks
```

```
        System.out.println("Enter
```

```
        int i = 0;
```

```
        while(i < 3)
```

```
            System.out.println("Enter
```

```
            marks[i]
```

```
            i++
```

```
        }
```

```
        Student s = new Student(roll, name, marks);
```

```
        students[count] = s;
```

```
        count++;
```

```
    } catch (Exception e) {
```

```
        System.out.println("Invalid input");
```

```
    } catch (Exception e) {
```

```
        System.out.println("Invalid input");
```

```
    } catch (Exception e) {
```

```
        System.out.println("Invalid input");
```

```
class ResultManager {
```

```
Scanner input = new Scanner(System.in);
```

```
Student[] Students = new Student[100];
```

```
int count = 0;
```

```
void addStudent() {
```

```
try {
```

```
    Sout("Enter Roll no ");
```

```
    int roll = input.nextInt();
```

```
    input.nextLine();
```

```
    Sout("Enter Student name");
```

```
    String name = input.nextLine();
```

```
    int[] marks = new int[3];
```

```
    Sout("Enter marks for 3 sub");
```

```
    int i = 0;
```

```
    while (i < 3) {
```

```
        Sout("Sub "+(i+1)+" : ");
```

```
        marks[i] = input.nextInt();
```

```
        i++;
```

```
    }
```

```
    Student st = new Student(roll, name, marks);
```

```
    st.validateMarks();
```

```
    Students[count++] = st;
```

```
    Sout("Student added");
```

```
} catch (InvalidMarksException e) {
```

```
    Sout("Error " + e.getMessage());
```

```
} catch (InputMismatchException e) {
```

```
    Sout("Error");
```

```

    } finally {
        sout("add student() operation completed.\n");
    }
}

public void showStudentDetails() {
    try {
        sout("Enter roll no to search");
        int roll = input.nextInt();
        int i = 0;
        boolean found = false;
        while(i < count) {
            Student s = Students[i];
            if(s != null && s.rollNumber == roll) {
                s.displayResult();
                found = true;
                break;
            }
            i++;
        }
        if(!found) {
            System.out.println("Student not found");
        }
    } catch (InputMismatchException e) {
        sout("Invalid Input");
        input.nextLine();
    }
}

public void mainMenu() {
    int choice = 0;
    try {
        do {
            sout("1: add student");

```

```

        cout << "Show details";
        cout << "Exit";
        cout << "Enter Choice";
        choice = input.nextInt();
        if (choice == 1) {
            addStudent();
        } else if (choice == 2) {
            showStudentDetails();
        } else if (choice == 3) {
            cout << "Exited successfully";
        } else {
            cout << "Invalid Option";
        }
    }
}

```

```

    } while (choice != 3);
}

```

```

} catch (InputMismatchException e) {
    cout << "Invalid Input";
} finally {
    cout << "Programme ended";
}
}

```

```

}

```

```

public static void main (String[] args) {
    ResultManager obj = new ResultManager();
    obj.mainMenu();
}
}

```